

Creating specialized classes for fire risk using machine learning methods

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Abstract

This study explores various machine learning techniques for assessing forest fire risk. The data used in this study have been selected from three different areas in Greece over the past 25 years (2000 - 2024). The first area of interest is Attica, the second area is eastern Macedonia and Thrace, and the third area is Crete. For each of the aforementioned areas, a separate data file was created, an analysis was carried out, and specialized definitions - classes were created for the forest fire risk occurrence. This study utilized a range of traditional machine learning techniques alongside novel methods based on Grammatical Evolution, including automated feature construction from existing attributes and the automatic generation of classification rules. For the Attica dataset, our results showed that GENCLASS achieved an error rate of 12.43%, corresponding to an accuracy of 87.57%. For the Eastern Macedonia & Thrace dataset, FC2 (MLP) reached an error rate of 11.87% and an accuracy of 88.13%, while GENCLASS performed very similarly with 11.93% error and 88.07% accuracy. Finally, for the Crete dataset, GENCLASS recorded an error rate of 15.11%, resulting in an accuracy of 84.89%. GENCLASS demonstrated stable performance across all three datasets, validating the climatological attributes and the definitions that underpin the construction of the Final_DangerClass_combination.

Keywords: Fire risk; Machine learning; neural networks

1. Introduction

This study presents a new rule for forest fire risk, which specifies and adapts the climatic and forest fire data with a new framework instead of the well known 30/30/30. In short the rule 30/30/30 means that there is a risk of fire when the following conditions prevail: temperatures >30, wind >30, and humidity <30. However, at this point we will briefly give a sample of analyses and experiments, with new definitions, which are specific to each region. For example, the dangerous mean temperature point for Attica is 23.6 °C or 73.7 °F, for Eastern Macedonia and Thrace is 20 °C or 68.1 °F, and for Crete is 20.17 °C or 70.1 °F. In other words, wildfires do not necessarily follow our predefined rules, however, our newly developed, region - with specific rules - adapted to the unique characteristics of each area, can contribute to more effective fire prevention and suppression efforts. In this context, it is important to highlight several real world case studies, that demonstrate the necessity of establishing new, region specific thresholds for wildfire ignition risk.

- A first example is the wildfire that broke out during the winter season, specifically in January 2025, in California, lasting 24 days [1]. Moreover, the study by Bouvier et al.

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(2025) indicates that an improved wildfire prevention model is required for periods traditionally considered as non hazardous for fire occurrence [2]. It is worth noting that, based on the meteorological data we retrieved from Visual Crossing for the region of California for the period 01/01/2025 - 31/01/2025, for this event, the mean temperature was approximately 5 °C (42 °F), with maximum temperatures reaching only 15 °C (60 °F)[3].

- A second example is the wildfire that broke out during the early spring season, specifically in March 2022, in South Korea [4]. Notably, by examining the temperature records for that specific period in the region of South Korea, the nighttime minimum temperatures dropped to -4 °C (24.8 °F), while daytime maximum temperatures ranged between 12 °C (53.6 °F) and 14 °C (57.2 °F) [5].
- A third example is the wildfire that occurred in May 2025 near the city of Chita in Russia [6]. During that period, temperatures were approximately 17.2 °C (63 °F), with the lowest recorded temperature reaching 4.4 °C (40 °F) [7].
- A fourth example is the winter wildfire that occurred in Ofunato, Japan, in February 2025 [8]. During that period, temperatures ranged from -1.6 °C (29 °F) to a maximum of 3.8 °C (39 °F) [9].

In summary, the aforementioned incidents represent only a subset of the wildfire events identified through the literature review, all of which fall outside the scope of the traditional 30/30/30 rule. This highlights the need to develop and examine new, region specific and context adapted metrics for assessing wildfire occurrence risk, based on meteorological variables and operational firefighting data.

In our search for studies related to our own work, we initially focused on identifying research that examined the application of the 30/30/30 rule and its reported outcomes, using the keywords: *30/30/30 rule fire risk*. The review of the retrieved studies indicated that only one verification oriented investigation of the rule has been conducted, and its findings were not satisfactory. Specifically, the study by Cuevas and Rodriguez (2021) highlights that the 30/30/30 rule does not account for the majority of large scale wildfires, explaining only about 10% of such events. In contrast, it appears more relevant for medium sized wildfires, accounting for approximately 66% of these cases. Moreover, temperature and relative humidity emerge as the most critical variables, as over 80% of the analyzed wildfires met these two conditions [10]. We will proceed by presenting studies and research works that examine the impact of meteorological variables and modeling approaches on forest fires, comparable to our own, and, where applicable, we will conclude by comparing their reported performance metrics with the findings of the present study.

The study by Cortez and Morais (2007) employed meteorological variables such as: temperature, relative humidity, rainfall, and wind, using an SVM model capable of predicting the burned area of small fires. Their approach achieved accurate predictions in 46% of the cases when an error of ≤ 1 ha was allowed, and in 61% of the cases when the admissible error increased to ≤ 2 ha [11]. The publication by Yunhao et al. (2007) focuses on comparing two wildfire risk indices: the Fire Potential Index (FPI) and the Integrated Fire Susceptibility Index (IFSI). Their experimental results, based on the Area Under the Curve (AUC) of the REC curves, showed that IFSI performed as the superior model with an accuracy of approximately 78%, whereas FPI achieved a prediction accuracy of about 73% [12].

Next on the research by Sakr et al. (2011) employed only two meteorological variables cumulative precipitation and relative humidity, together with two artificial intelligence techniques, artificial neural networks (ANN) and support vector machines (SVM), to accurately predict the risk of fire occurrence. Their results showed that, for the SVM model, each month included at least one iteration in which the testing accuracy exceeded 97%

Table 1. Comparative overview of accuracy/performance results reported by previous studies on forest fire risk prediction, as identified in the literature review.

Authors names	Results
Diaz Cuevas, M. D. P., & Limones Rodriguez, N.[10]	10% accuracy (large fires) & 66% (medium fires)
Cortez, P., & Morais, A. D. J. R[11]	46% error (of ≤ 1ha) & 61% (≤ 2 ha)
Yunhao, C., Jing, L., & Guangxiong, P.[12]	IFSI accuracy (78%) & FPI (73%)
Sakr, G. E., Elhajj, I. H., & Mitri, G.[13]	SVM accuracy (97%)
Agarwal, P., Tang, J., Narayanan, A. N. L., & Zhuang, J[14]	Gradient Boosting Tree accuracy (93.3%) & test set (97%)
Masinda, M. M., Li, F., Qi, L., Sun, L., & Hu, T[15]	FWI Wilcoxon test accuracy (72.7%)
Karali, A., Varotsos, K. V., Giannakopoulos, C., Nastos, P. P., & Hatzaki, M[16].	FWI & ISI accuracy (60%)
Parvar, Z., Saeidi, S., & Mirkarimi, S.[17]	AUC value of 0.83

[13]. Analogous result were reported in the study by Agarwal et al. (2020) who applied the Gradient Boosting Tree algorithm using data from the National Fire Incident Reporting System (NFIRS) together with meteorological information from the National Oceanic and Atmospheric Administration (NOAA) to predict economic losses caused by fire incidents. Their results demonstrated strong predictive performance, achieving 93.3% accuracy on the full dataset and 97% accuracy on the test set [14]. The subsequent field study by Masinda et al. (2022) examined the daily assessment of wildfire danger in Northeastern China using the Canadian Forest Fire Weather Index (FWI). Their results indicated that the Canadian FWI performed well, with the Wilcoxon paired sample test showing satisfactory agreement in 72.7% of the cases [15]. The same modeling framework, combining the Canadian Forest Fire Weather Index (FWI) with the Initial Spread Index (ISI), was examined by Karali et al. (2023) in their assessment of hazardous wildfire prone conditions. Their findings indicated that lead time plays a decisive role in forecast performance, with the probability of correct prediction reaching approximately 60% [16]. In their study, Parvar et al. (2024) developed a wildfire risk map by integrating meteorological variables such as: Land Surface Temperature (LST), precipitation, and historical fire occurrence data. The performance of their model was statistically evaluated using the ROC curve, achieving an AUC value of 0.83%, indicating strong predictive capability [17].

At this point, we will present, in a table 1, the results of the aforementioned studies in order to enable a direct comparison among them Table 1. In Section4 a concise comparison will also be provided between the findings of each related study and those of our own research.

The remaining of this paper has the following structure: the Section 2 presents the used dataset and the new threshold values for wildfire occurrence, the Section 3 illustrates the experimental results finally the Section 4 It provides a concise presentation of the results, accompanied by a comparative discussion with the previously cited studies in the table 1.

2. Materials and Methods

In this section, a detailed presentation of the datasets used as well as the machine learning techniques used in the experiments performed will be provided.

Figure 1. Map of Greece highlighting the three geographic study areas (Study Area 1: Attica, Study Area 2: Eastern Macedonia & Thrace, Study Area 3: Crete)



2.1. The dataset employed

For the purposes of our study, we combined two distinct data sources. On the one hand, to analyze the climatological conditions, we employed Historical Weather Data from Visual Crossing <https://www.visualcrossing.com/> [3] (last accessed on 02 February 2026), which compiles information from a range of authoritative sources, including national meteorological agencies, global climate datasets, and satellite observations. Access to the historical dataset required completing the registration process, after which we selected the data-access interface, specified the region of interest, and defined the temporal range. Data retrieval was then carried out under the free access tier, which allows up to 1,000 records per day. The procedure is generally straightforward, with the only notable constraint being the daily limit of 1,000 data points.

On the other hand, for the analysis of forest / wild fires, we relied on the open data provided by the Hellenic Fire Service https://www.fireservice.gr/el_GR/synoladedomenon (last accessed on 02 February 2026). The fire data are collected and updated each year by the individual fire stations responsible for reporting incidents.

2.2. The dataset description

For this study, we obtained historical meteorological data for the period 2000 – 2024, providing a roughly 25 year record to support our fire duration according to climate patterns analysis. We additionally selected regions across different latitudes and longitudes, in Greece, to achieve a more comprehensive representation of the study areas. The selected regions of interest are: Attica, Eastern Macedonia and Thrace, and Crete 1.

The climatic and fire dataset for all regions covers the period from January 1, 2000, to December 31, 2024. Specifically, Attica provides 9.131 climatic dataset and 2.736 fire records, Eastern Macedonia and Thrace provides 9.131 climatological records and 10.358 fire records, and finally Crete provides 9.121 climatological records and 6.036 fire record. The variation in the number of climatological records is attributable to gaps (missing values) in the data retrieved from Visual Crossing, which did not influence our results. Before proceeding with the merging of the two data files, the experiments focused exclusively on forest fire incidents in order to maintain homogeneity in our dataset with respect to fire

behavior characteristics. In other words, we chose to proceed with the following categories : (9) Burned forests, (10) Burned forest area, and (11) Burned grove (small forest) Table 3. Following this refinement, Attica provided 1.859 forest fire records, Eastern Macedonia and Thrace 2.653, and Crete 2.789.

Afterwards, the climatological dataset merged with the forest fire dataset, retaining only the climate records corresponding to the fire ignition date. The merging process was performed combining the climatological data Table 2 with the fire incident data Table 3 based on the date field.

Table 2. Primary table of raw climatological variables (33 attributes) retrieved from the Visual Crossing platform for the period 2000–2024.

Number	Data Description	Values
1	Region	Area
2	Date	Year/month/day
3	Tempmax (Fahrenheit)	°F
4	Tempmin (Fahrenheit)	°F
5	Temp (Fahrenheit) daily mean average	°F
6	Feelslikemax (Fahrenheit)	°F
7	Feelslikemin (Fahrenheit)	°F
8	Feelslike (Fahrenheit)	°F
9	Dew	°F
10	Humidity	%
11	Precipitation	in
12	Precipprob	%
13	Precipcover	%
14	Preciptype	Type(s)
15	Snow	in
16	Snowdepth	in
17	Windgust (miles)	mph
18	Windspeed (miles)	mph
19	Winddir	(°)
20	Sealevel	mb
21	Cloudcover	%
22	Visibility	mi
23	Solarradiation	(W/m²)
24	Solarenergy	(MJ/m²)
25	Uvindex	(0-10)
26	Severisk	(0-100)
27	Sunrise	time
28	Sunset	time
29	Moonphase	(0-1)
30	Conditions	general observations
31	Descriptions	general observations day/night
32	Icon	summarized observations
33	Stations	Code for stations

Table 3. Primary table of raw operational forest fire incident data (16 attributes) as recorded by the Hellenic Fire Service.

Number	Data Description
1	Fire department
2	Forest department
3	Location
4	Region
5	Date of fire ignition
6	Time of fire ignition
7	Extinguish date
8	Extinguish time
9	Burned forests
10	Burned forests area
11	Burned grove (small forest)
12	Burned grassland
13	Burned swamps
14	Burned agricultural
15	Burned crops
16	Burned garbage dumps

Accordingly, we created three new datasets, one for each study region. Each region was examined and analyzed independently, resulting in a distinct fire risk classification for every area. Consequently, the variables processed in order to construct the class representing danger forest fire ignition, for all three datasets were the following: mean daily temperature, daily humidity values, precipitation occurrence, wind gusts, wind speed, wind direction, time of fire ignition, month of fire ignition, and fire duration, with the final output expressed as the Final_DangerClass_combination Table 4. In the end, in order to ensure impartiality in our experiments, we removed the FireDuration feature, since the model would otherwise be able to see the answer, as the accuracy for all datasets reached 100%.

Table 4. Final derived variables/classes obtained after processing the raw data, used to construct the target variable Final_DangerClass_combination.

Number	Data
1	TempSpecialClass (daily mean average)
2	HumiditySpecialClass
3	PrecipitationBasicClass
4	WindgustSpecialClass
5	WindspeedSpecialClass
6	WinddirSpecialClass
7	TimeClass
8	Monthclass
9	Fireduration (Removed)
10	Final_DangerClass_combination

2.2.1. Attica area and data description

The regional unit of Attica covers an area of 3,808.1 km², with Athens as its capital. Between 2001 and 2025, Attica experienced a loss of 22 kha (220 km²) of tree cover attributable to wildfires, along with an additional 3.3 kha (33 km²) lost due to other disturbance factors. The most severe year within this period was 2021, during which fires accounted for 8.5 kha (85 km²) of tree cover loss, representing 99% of the total loss recorded that year, according to the global forest watch [18].

A brief description of the Attica dataset follows. The final dataset consisted of 1.859 records and 8 attributes, with the last attribute representing the target class, namely: Final_DangerClass_combination. The classes followed a binary scheme, where 0 denoted a non hazardous value, and 1 denoted as a hazardous value, in the following Table 5.

Table 5. Definition of the threshold values (temperature, humidity, wind, wind direction, ignition time, month) separating the non-hazardous (Class 0) from the hazardous (Class 1) fire danger class for the Attica region, with the corresponding number of records per class.

Features	Class 0 (non hazard)	Class 1 (hazard)
TempSpecialClass (daily mean average)	$\leq (73.6^{\circ}\text{F})$ or	$73.7^{\circ}\text{F} = > (\text{or } 23.6^{\circ}\text{C})$
HumiditySpecialClass	$49.4\% = >$	$< = 49.3\%$
PrecipitationBasicClass	rain	no rain
WindgustSpecialClass	$< = 10\text{miles}$	$10.01\text{miles} = >$
WindspeedSpecialClass	$< = 16\text{miles}$	$16.1\text{miles} = >$
WinddirSpecialClass	(S, SE, E, W)	(SW, N, NE, NW)
TimeClass	$< = 12:30$	$12:31 = >$
Monthclass	(1,2,3,4,5,9,10,11,12)	(6,7,8*)
Final_DangerClass_combination	844 data	1.015 data

In summary, the findings of the study identified new threshold values for wildfire occurrence in the region of Attica, as follows: temperature $\geq 73.7^{\circ}\text{F}$, humidity $\leq 49.3\%$, 0% precipitation, wind gust speed ≥ 10.01 mph, wind speed ≥ 16.1 mph, wind direction in the sectors SW, N, NE, and NW, ignition time from 12:31 on wards, and hazardous months corresponding to June, July, and August*.

2.2.2. Eastern Macedonia and Thrace area and data description

The regional unit of Eastern Macedonia and Thrace, covers an area of 14.158 km², with Komotini as its capital. Between 2001 to 2025, Eastern Macedonia and Thrace experienced a loss of 40 kha (400 km²) of tree cover attributable to wildfires, along with an additional 11 kha (110 km²) lost due to other disturbance factors. The most severe year within this period was 2023, during which fires accounted for 31 kha (310 km²) of tree cover loss, representing 94% of the total loss recorded that year, according to the global forest watch[18].

A brief description of the Eastern Macedonia and Thrace dataset follows. The final dataset consisted of 2.653 records and 8 attributes, with the last attribute representing the target class, namely: Final_DangerClass_combination. The classes follow the same binary scheme as previously. A concise summary table of these variables is presented below Table 6.

Table 6. Definition of the threshold values (temperature, humidity, wind, wind direction, ignition time, month) separating the non-hazardous (Class 0) from the hazardous (Class 1) fire danger class for the Eastern Macedonia & Thrace region, with the corresponding number of records per class.

Features	Class 0 (non hazard)	Class 1 (hazard)
TempSpecialClass (daily mean average)	$< = 68^{\circ}\text{F}$	$68.1^{\circ}\text{F} = > (\text{or } 20^{\circ}\text{C})$
HumiditySpecialClass	$64.1\% = >$	$< = 64\%$
PrecipitationBasicClass	rain	no rain
WindgustSpecialClass	$< = 7\text{miles}$	$7.1\text{miles} = >$
WindspeedSpecialClass	$< = 6\text{miles}$	$6.1\text{miles} = >$
WinddirSpecialClass	(W,NW,N)	(E,SE,SW,S,NE)
TimeClass	$< = 11:30$	$11:31 = >$
Monthclass	(1,2,3,4,9,11,12)	(5,6,7,8,10*)
Final_DangerClass_combination	1.093	1.559

In summary, the findings of the study identified new threshold values for wildfire occurrence in the region of Eastern Macedonia and Thrace, as follows: temperature $\geq 68.1^\circ\text{F}$, humidity $\leq 64.1\%$, 0% precipitation, wind gust speed ≥ 7.1 mph, wind speed ≥ 6.1 mph, wind direction in the sectors E, S, SW, SE, and NE, ignition time from 11:31 on wards, and hazardous months corresponding to May, June, July, August and October.

2.2.3. Crete area and data description

The regional unit of Crete covers an area of 8.450 km², with Herakleio as its capital. Between 2001 to 2025 Crete experienced a loss of 1.3 kha (13 km²) of tree cover attributable to wildfires, along with an additional 3.2 kha (33 km²) lost due to other disturbance factors. The most severe year within this period was 2008 during which fires accounted for 210 ha (2.1 km²) of tree cover loss, representing 44%, of the total loss recorded that year, according to the global forest watch [18].

A brief description of the Crete datasets follows. The final dataset consisted 2.789 records and 8 attributes, with the last attribute representing the target class, namely: Final_DangerClass_combination Table 7.

Table 7. Definition of the threshold values (temperature, humidity, wind, wind direction, ignition time, month) separating the non-hazardous (Class 0) from the hazardous (Class 1) fire danger class for the Crete region, with the corresponding number of records per class.

Features	Class 0 (non hazard)	Class 1 (hazard)
TempSpecialClass (daily mean average)	$\leq 70^\circ\text{F}$	$70.1^\circ\text{F} = > (\text{or } 20.17^\circ\text{C})$
HumiditySpecialClass	$66.1\% = >$	$\leq 66\%$
PrecipitationBasicClass	rain	no rain
WindgustSpecialClass	$\leq 9\text{miles}$	$9.1\text{miles} = >$
WindspeedSpecialClass	$\leq 16\text{miles}$	$16.1\text{miles} = >$
WinddirSpecialClass	(S, SW, SE, E, W)	(N, NE, NW)
TimeClass	$\leq 12:30$	$12:31 = >$
Monthclass	(1,2,3,4,5,9,10,11,12)	(6,7,8*)
Final_DangerClass_combination	1.383	1.404

In summary, the findings of the study identified new threshold values for wildfire occurrence in the region of Crete, as follows: temperature $\geq 70.1^\circ\text{F}$, humidity $\leq 66\%$, 0% precipitation, wind gust speed ≥ 9.1 mph, wind speed ≥ 16.1 mph, wind direction in the sectors N, NE, and NW, ignition time from 12:31 on wards, and hazardous months corresponding to June, July, and August.

3. Results

For the purposes of our experiments, we employed the available optimization package Optimus[19], which can be downloaded from <https://github.com/itsoulos/GlobalOptimus> (accessed on 20 June 2026). For the feature construction process used in the FC2(RF) and FC2(MLP) algorithms, we utilized the freely available programming tool QFc[20], accessible at <https://github.com/itsoulos/QFc> (accessed on 20 June 2026). In addition, the GENCLASS [21] system was employed in our experiments. GENCLASS is a free software framework entirely written in ANSI C++, available by <https://github.com/itsoulos/GenClass/> (accessed on 20 June 2026) designed to construct classification programs in a C like programming language for the purpose of classifying input data. It operates as an evolutionary rule induction method, generating interpretable decision rules through genetic search mechanisms.

Each experiment was executed 30 times, using different seeds for the random number generator in every run. In addition, the ten-fold cross-validation procedure was applied to

ensure robust validation of the experimental results. The proposed feature construction method was applied independently to each fold, and the average classification errors on the corresponding test sets were computed.

The results demonstrated that the model GENCLASS achieved the lower error accuracy in predicting the final danger class, indicating that the newly developed fire danger rule is fully identifiable and predictable based on the combined climatological and operational fire incident data. . In addition, the FC2 (MLP) algorithm achieved the second best performance for the datasets: Attica Table 8, and for Crete region Table 10 as for Eastern Macedonia – Thrace the FC2 (MLP) model achieved marginally better accuracy by 0.06% Table 9.

In the following, tables and graphs present our results for each region regarding the classification error. In essence, the classification error indicates that the lower the reported percentage, the stronger and more accurate the model's performance.

3.1. Attica Class Error

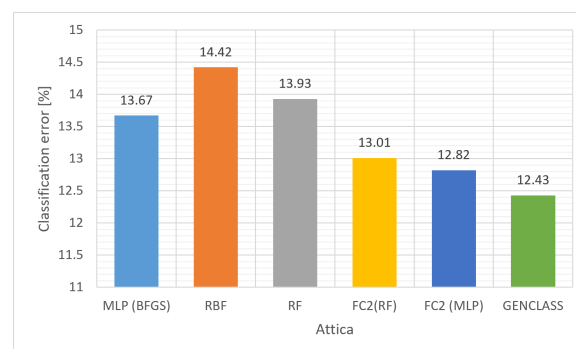
Attica contains 1.859 records, of which 844 correspond to class 0 and 1.015 to class 1. The model GENCLASS achieved 12.43% error accuracy, thus achieving a correct accuracy rate of 87.57% in predicting the final danger class. In addition, the FC2 (MLP) algorithm achieved the second best performance with 12.82% error, Table 8 & Figure 2. Conversely, the RBF algorithm shows noticeably lower performance, with an error rate of 14.42%.

Table 8. Classification error rates (%) achieved by each machine learning technique (MLP, RBF, RF, FC2(RF), FC2(MLP), GENCLASS) across the 10 cross-validation folds and their average - Attica region.

	MLP (BFGS)	RBF	RF	FC2(RF)	FC2 (MLP)*	GENCLASS
1o set	11.83%	13.44%	12.90%	14.05%	13.51%	12.97%
2o set	11.83%	9.68%	12.90%	9.73%	7.95%	9.73%
3o set	12.37%	13.98%	15.05%	13.51%	12.22%	12.43%
4o set	15.05%	14.52%	16.13%	15.14%	15.19%	15.14%
5o set	13.98%	19.35%	13.44%	11.35%	9.95%	8.65%
6o set	13.44%	14.52%	13.44%	14.05%	14.38%	14.05%
7o set	15.05%	13.98%	16.13%	15.14%	15.62%	15.14%
8o set	11.83%	14.52%	13.44%	11.29%	12.85%	11.89%
9o set	11.29%	11.29%	10.75%	11.83%	12.31%	14.05%
10o set	20%	18.92%	15.14%	14.05%	14.22%	10.27%
Average	13.67%	14.42%	13.93%	13.01%	12.82%	12.43%

For reasons of brevity, FC2 (MLP) is denoted as FC2 in the graph.

Figure 2. Bar chart of the average classification error rate (%) for each machine learning technique - Attica region.



3.2. Eastern Macedonia & Thrace Class Error

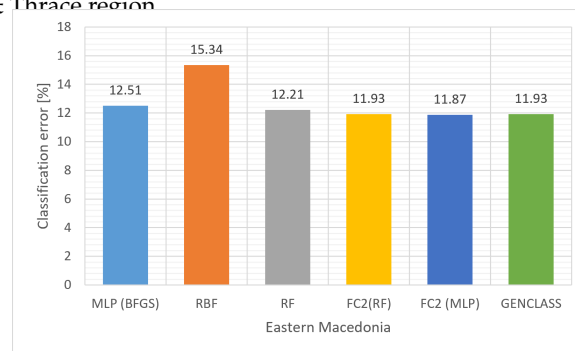
Eastern Macedonia & Thrace contains 2.653 records, of which 1.093 correspond to class 0 and 1.559 to class 1. The model FC2 (MLP) achieved 11.87% error accuracy, thus achieving a correct accuracy rate of 88.13% in predicting the final danger class. In addition, the GENCLASS along with FC2(RF) achieved the second best performance with 11.93% error, Table 9 & Figure 3. On the other hand, the RBF algorithm shows noticeably lower performance, with an error rate of 15.34%.

Table 9. Classification error rates (%) achieved by each machine learning technique (MLP, RBF, RF, FC2(RF), FC2(MLP), GENCLASS) across the 10 cross-validation folds and their average - Eastern Macedonia & Thrace region.

	MLP (BFGS)	RBF	RF	FC2(RF)	FC2 (MLP)*	GENCLASS
1o set	15.41%	15.04%	14.66%	10.57%	11.02%	10.57%
2o set	10.53%	15.79%	10.90%	13.26%	13.47%	13.96%
3o set	11.28%	11.65%	10.90%	12.45%	12.68%	11.32%
4o set	15.09%	16.60%	14.34%	13.71%	10.42%	11.32%
5o set	13.58%	17.36%	13.96%	11.32%	12.00%	11.70%
6o set	13.58%	15.09%	12.45%	10.57%	10.57%	10.19%
7o set	10.94%	15.47%	9.06%	11.32%	11.62%	12.83%
8o set	11.70%	16.23%	12.08%	14.96%	14.34%	13.21%
9o set	15.85%	15.85%	16.60%	9.43%	9.28%	11.70%
10o set	7.17%	14.34%	7.17%	11.70%	13.32%	12.45%
Average	12.51%	15.34%	12.21%	11.93%	11.87%	11.93%

For reasons of brevity, FC2 (MLP) is denoted as FC2 in the graph.

Figure 3. Bar chart of the average classification error rate (%) for each machine learning technique - Eastern Macedonia & Thrace region



3.3. Crete Class Error

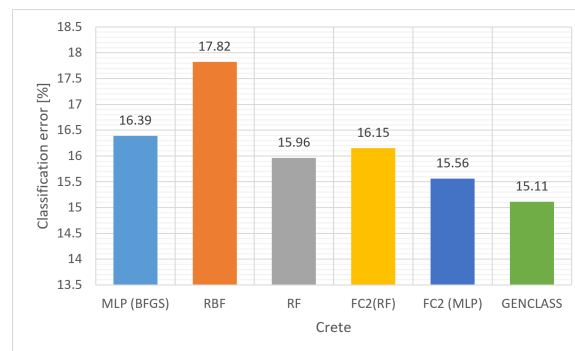
Crete contains 2.789 records, of which 1.383 correspond to class 0 and 1.404 to class 1. The model GENCLASS achieved 15.11% error accuracy, thus achieving a correct accuracy rate of 84.89% in predicting the final danger class. In addition, the FC2 (MLP) algorithm achieved the second best performance with 15.56% error, Table 10 & Figure 4. Instead, the RBF algorithm shows noticeably lower performance, with an error rate of 17.82%.

Table 10. Classification error rates (%) achieved by each machine learning technique (MLP, RBF, RF, FC2(RF), FC2(MLP), GENCLASS) across the 10 cross-validation folds and their average - Crete region.

	MLP (BFGS)	RBF	RF	FC2(RF)	FC2 (MLP)*	GENCLASS
1o set	16.49%	21.51%	16.49%	15.11%	15.29%	14.75%
2o set	16.85%	16.85%	17.56%	14.75%	14.75%	14.39%
3o set	16.13%	17.20%	17.20%	15.83%	14.64%	14.03%
4o set	18.64%	19.71%	19%	16.55%	15.76%	16.55%
5o set	17.20%	19%	16.85%	14.03%	14.06%	14.75%
6o set	14.70%	15.41%	13.62%	19.78%	15.90%	15.47%
7o set	16.49%	16.49%	13.62%	18.35%	15.90%	15.47%
8o set	18.64%	18.28%	13.98%	14.39%	14.71%	14.75%
9o set	16.85%	17.56%	17.92%	16.91%	16.33%	13.31%
10o set	11.87%	16.19%	13.31%	15.83%	15.50%	14.75%
Average	16.39%	17.82%	15.96%	16.15%	15.56%	15.11%

For reasons of brevity, FC2 (MLP) is denoted as FC2 in the graph.

Figure 4. Bar chart of the average classification error rate (%) for each machine learning technique - Crete region.



In summary, the results show that GENCLASS achieved better overall performance compared to the other methods and consistently remained very close to FC2 (MLP), with the only exception being the Eastern Macedonia & Thrace dataset, where FC2 (MLP) performed marginally better by 0.06%.

The subsequent section provides the report and detailed description of the precision and recall metrics.

3.4. Precision and recall values.

Precision and recall are evaluation metrics used to assess the performance of a machine learning model. Precision expresses the proportion of predicted positive cases that are actually correct, while recall indicates the proportion of true positive cases that the model successfully identifies. Together, these metrics provide essential insight into the effectiveness of a model in classification tasks.

Precision is calculated as:

$$\text{Precision} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalsePositives}}$$

Precision reflects the model's ability to avoid false positives, making it a key indicator of prediction reliability. In other words, the higher the value, the fewer the errors.

The Recall value confirms the model's ability to correctly identify the positive class, reflecting how many of the actual positive instances were successfully detected. As with precision and recall, the higher the value, the more successful the classification

Recall is calculated as:

$$\text{Recall} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalseNegatives}}$$

3.5. Attica precision and recall.

The precision and recall results for Attica show that all techniques deliver high performance in determining the final danger class. In fact, the values ranged between 85.32% and 88.50%. Among these methods, GenClass, FC2(RF), RF, and FC2(MLP) demonstrated the strongest performance 5, 6. In this comparison as well, RBF holds the lowest performance, which stands at approx 85%.

For reasons of brevity, FC2 (MLP) is denoted as FC2 in the graph.

Figure 5. Bar chart of precision values (%) for each machine learning technique - Attica region.

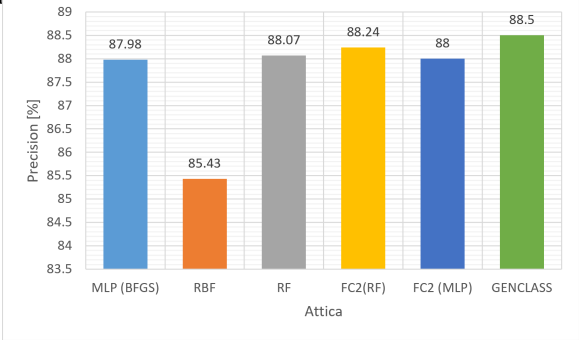
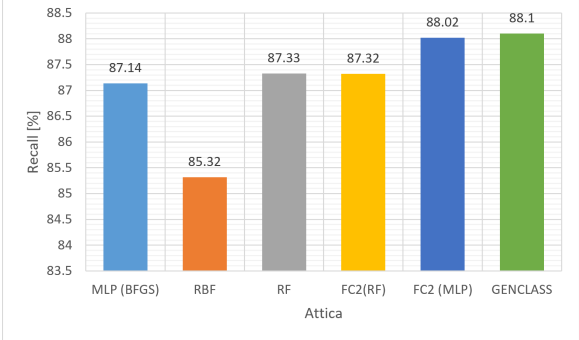


Figure 6. Bar chart of recall values (%) for each machine learning technique - Attica region.



3.6. Eastern Macedonia & Thrace precision and recall.

The data for Eastern Macedonia & Thrace exhibit even higher performance levels, ranging from 83.82% to 89.45%. The techniques that stand out are: RF, GENCLASS, MLP(BFGS), FC2(RF) and FC2(MLP). 7, 8. Also here the RBF holds the lowest performance, which stands at approx 83%.

For reasons of brevity, FC2 (MLP) is denoted as FC2 in the graph.

Figure 7. Bar chart of precision values (%) for each machine learning technique - Eastern Macedonia & Thrace region.

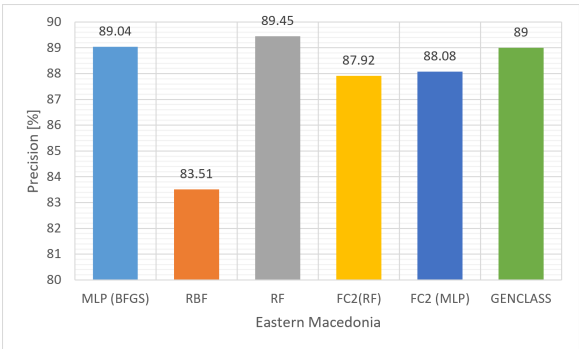
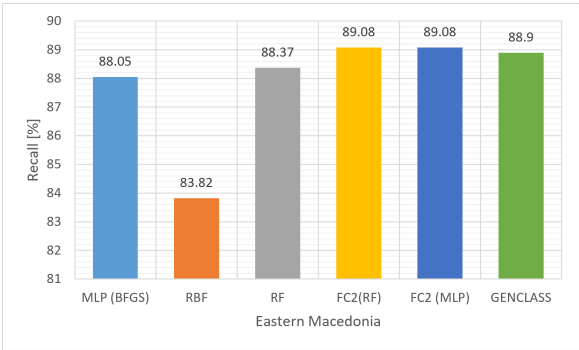


Figure 8. Bar chart of recall values (%) for each machine learning technique - Eastern Macedonia & Thrace region.



3.7. Crete precision and recall

Although Crete presents satisfactory values, its performance levels were lower compared to the other datasets. The results ranged from 82.09% up to 85.50%, and once again the techniques that stood out were: GENCLASS, RF, FC2(RF), and FC2(MLP). 9, 10. In addition, the RBF holds the lowest performance, which stands at approx 82%.

For reasons of brevity, FC2 (MLP) is denoted as FC2 in the graph.

Figure 9. Bar chart of precision values (%) for each machine learning technique - Crete region.

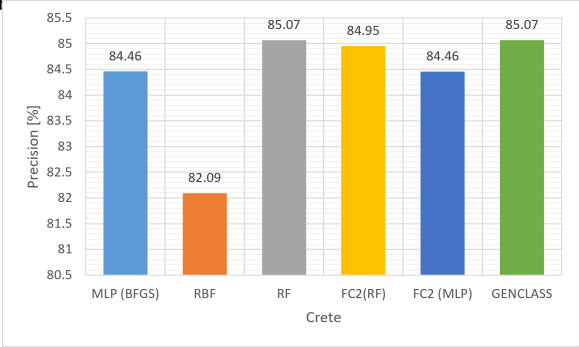
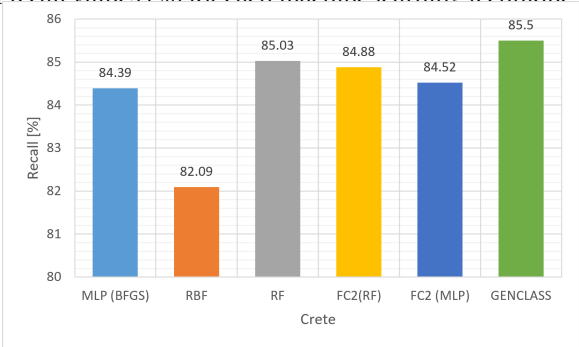


Figure 10. Bar chart of recall values (%) for each machine learning technique - Crete region.



The following table presents the average values of the precision and recall measures for each dataset and for every method Table 11. In this comparison as well, we observe the consistent performance of all methods. The best values are achieved by RF, FC2(RF), FC2 (MLP), and GENCLASS, while for the Eastern Macedonia & Thrace dataset, MLP (BFGS) stands out. Finally, the lowest performance is recorded by RBF.

Table 11. Comparative summary of average precision and recall values (%) for each machine learning technique, by study region (Attica, Eastern Macedonia & Thrace, Crete).

		MLP (BFGS)	RBF	RF	FC2 (RF)	FC2 (MLP)	GENCLASS
Attica	Precision	87.98%	85.43%	88.07%	88.24%	88%	88.50%
	Recall	87.14%	85.32%	87.33%	87.32%	88.02%	88.10%
Eastern Macedonia	Precision	89.04%	83.51%	89.45%	87.92%	88.08%	89%
	Recall	88.05%	83.82%	88.37%	89.08%	89.08%	88.90%
Crete	Precision	84.46%	82.09%	85.07%	84.95%	84.46%	85.07%
	Recall	84.39%	82.09%	85.03%	84.88%	84.52%	85.50%

4. Conclusions

This research developed new wildfire danger rules following an in depth examination of the environmental characteristics associated with each recorded fire incident and the corresponding analytical outcomes. For the purposes of the study, we employed several machine learning approaches, including Random Forest (RF), a set of feature construction techniques such as FC2 (RF) and FC2 (MLP), along with artificial Neural Network like RBF and MLP (BFGS) and finally GENCLASS a portable tool for data classification based on Grammatical Evolution.

For the completion of the project, we used climatological and operational firefighting data from 2000 to 2024 for three different regions of Greece, in order to highlight the distinct climatological conditions characterizing each of them. This perspective underscores that every region is unique and must be investigated according to its own climatological particularities.

Our results indicated stable performance across all datasets. Specifically, for the Attica dataset, GENCLASS achieved the best performance with the lowest error rate of 12.43%. It was followed by the feature construction techniques FC2 (RF) and FC2 (MLP), the artificial neural network MLP (BFGS), the Random Forest (RF), while RBF ranked last in terms of accuracy. In the precision and recall values, GENCLASS achieved the highest performance, ranging between 88.50% and 88.10%, respectively, while RBF ranked last with corresponding values of 85.43% and 85.32%.

The dataset for Eastern Macedonia & Thrace showed that the feature construction technique FC2 (MLP) achieved the best performance, with an error rate of 11.87%, outperforming the other methods by a small margin of 0.06%. In second place, GENCLASS and the feature construction technique FC2 (RF) tied, both recording an error rate of 11.93%. The artificial neural network RBF ranked last in this dataset, yielding an error rate of 15.34%. In the precision values, Random Forest (RF) achieved the best performance with 89.45%, followed by the artificial neural network MLP (BFGS) with 89.04%, and GENCLASS with 89%. RBF ranked last, yielding a precision of 83.51%. Regarding recall, GENCLASS achieved the second highest value at 88.90%. In second place, FC2 (RF) and FC2 (MLP) tied, both reaching 89.08%. Once again, RBF exhibited the lowest performance, with a recall value of 83.82%.

For the Crete dataset, GENCLASS achieved the best error accuracy with 15.11%, followed by FC2 (MLP) with 15.56%, while RBF ranked last with an error rate of 17.82%. In the precision and recall values, GENCLASS obtained the highest performance, ranging between 85.07% and 85.50%. In terms of precision, GENCLASS tied with Random Forest (RF) at 85.07%, whereas in recall RF ranked second with 85.03%. Finally, the RBF technique recorded the lowest performance in both precision and recall, with values of 82.09%.

At this stage, we compare our findings with the results reported in the previously cited studies from Table 1.

In 2021, the study that conducted from [10] indicated that the 30/30/30 rule does not provide accuracy, as it is overly general, and their experiments showed low accuracy rates ranging from 10% to 66%. Overall, our research is largely consistent with theirs, for this reason, we processed the climatological factors of each region and accordingly developed specialized new rules to replace the general 30/30/30 rule. The next paper [11] reported results within a similar range, achieving accuracy levels between 46% and 61%, while using four factors: temperature, humidity, rainfall, and wind, for the purposes of their experiments. The difference in our study is that we achieved better results, with an average error rate of 13.16% across the three datasets, corresponding to a correct prediction rate of mean average at 86.84%. In addition, our experiments were conducted using the following factors: temperature, humidity, rainfall, wind gusts, wind speed, wind direction, time of occurrence, and month. Continuing, the research by [12] compared two wildfire models, the IFSI and the FPI, reporting accuracy levels of 78% and 73%, respectively. The difference compared to our work lies both in our superior results and in the application of our own customized rules. The study by [13] took a different direction, focusing on only two features: humidity and precipitation, and achieved high accuracy levels of 97%. The main difference compared to our work lies in the number of features used, as we focused on eight features, with our accuracy averaging 86.84%. Although the following aim [14] combined operational wildfire data with meteorological data, it focused on estimating the economic damage caused by fires, achieving very strong results ranging from 93.3% to 97%. While the core structure of the data, namely the integration of operational and meteorological information, aligns with our own approach, the fundamental discrepancy lies in the fact that their work targeted economic loss estimation, whereas our research focused on deriving insights that would be operationally useful for wildfire suppression services. The subsequent researchers [15] attempted to verify the effectiveness of the Canadian FWI model in a region of China, reporting an accuracy of 72.7%. The distinction from our study lies in the fact that we developed our own “model”, tailored to the meteorological characteristics of each region, with better results. Using the same model, the next project from [16] on wildfire prediction for the region of Attica, where their results reached 60%. This comparison is particularly interesting, as they worked on one of the regions we also analyzed. Our results for Attica reached 87.57% accuracy and 12.43% error. Since as previously noted we did not rely on external models or generic rules, but instead developed our own region specific rules. The final study [17] combined meteorological data with operational fire data, focusing specifically on land surface temperature and precipitation, achieving high performance with a ROC value reaching 0.83. Although the core idea and the data integration approach are similar to ours, we employed a larger set of meteorological variables, achieving correspondingly high accuracy levels.

As shown in the comparison above, our study employed a larger set of environmental variables and additionally developed new rules for the emergence of the Final_DangerClass_combination, tailored to each region. Our results were validated through multiple machine learning approaches, yielding very good error rate performance for each dataset individually.

In summary, the findings of the study identified new threshold values for wildfire occurrence in the region of Attica, as follows: temperature $\geq 73.7^{\circ}\text{F}$, humidity $\leq 49.3\%$, 0% precipitation, wind gust speed ≥ 10.01 mph, wind speed ≥ 16.1 mph, wind direction in the sectors SW, N, NE, and NW, ignition time from 12:31 on wards, and hazardous months corresponding to June, July, and August. The accuracy of these values is further validated through the applied validation techniques, with GENCLASS achieving the best performance at 12.43% error, as presented in the Table 8 indicating a correct verification rate of 87.57%. Similarly, for the region of Eastern Macedonia and Thrace, the study identified

new threshold values for wildfire occurrence as follows: temperature $\geq 68.1^{\circ}\text{F}$, humidity $\leq 64.1\%$, 0% precipitation, wind-gust speed ≥ 7.1 mph, wind speed ≥ 6.1 mph, wind direction within the sectors E, S, SW, SE, and NE, ignition time from 11:31 onwards, and hazardous months corresponding to May, June, July, August, and October. Furthermore, the accuracy of these values is validated through the applied validation techniques, with FC2 (MLP) achieving the best performance at 11.87% error, as presented in the Table 9 indicating a correct accuracy rate of 88.13%. In the same way, for the region of Crete, the study identified new threshold values for wildfire occurrence as follows: temperature $\geq 70.1^{\circ}\text{F}$, humidity $\leq 66\%$, 0% precipitation, wind-gust speed ≥ 9.1 mph, wind speed ≥ 16.1 mph, wind direction within the sectors N, NE, and NW, ignition time from 12:31 onwards, and hazardous months corresponding to June, July, and August. Here as well, the newly derived values are validated by the performance of the applied techniques, with GENCLASS achieving the best result at 15.11% error, as presented in the Table 10 indicating a correct verification rate of 84.89%.

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